ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

N-GRAM ESTIMATION

UC San Diego

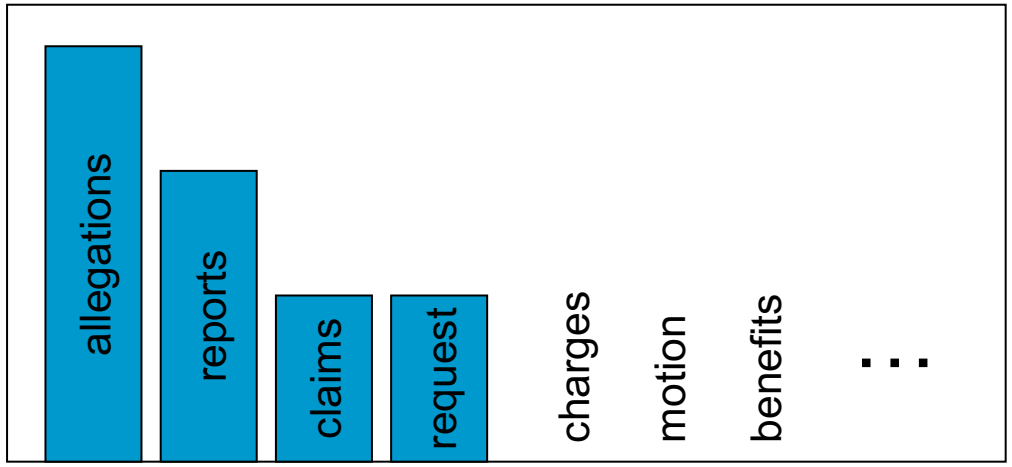
COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

N-Gram Estimation

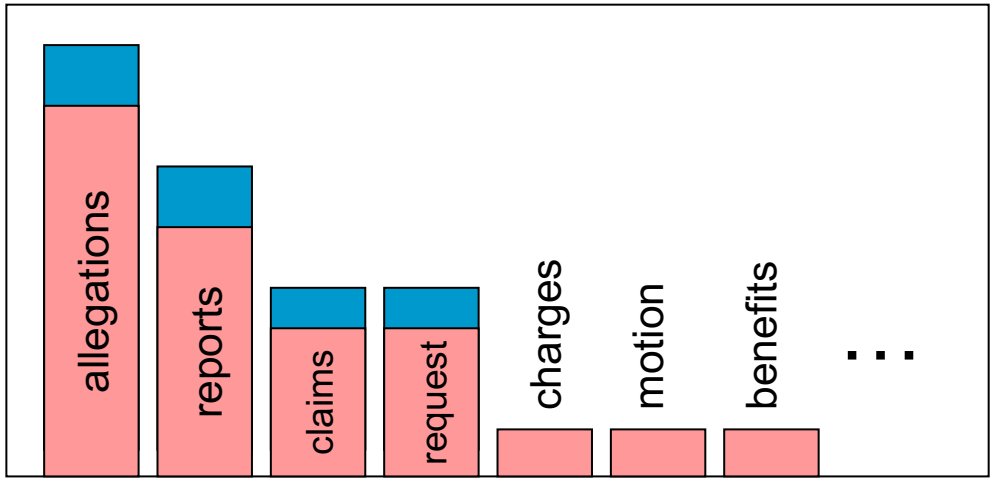
We often want to make estimates from sparse statistics:

P(w | denied the)
3 allegations
2 reports
1 claims
1 request
7 total



Smoothing flattens spiky distributions so they generalize better:

P(w | denied the)
2.5 allegations
1.5 reports
0.5 claims
0.5 request
2 other
7 total



Very important, but easy to do badly.

How do we measure LM "goodness"?

- Shannon's game: predict the next word

When I eat pizza, I wipe off the _____

grease 0.5
sauce 0.4
dust 0.05
….
mice 0.0001
….
the 1e-100

Formally: define test set (log) likelihood

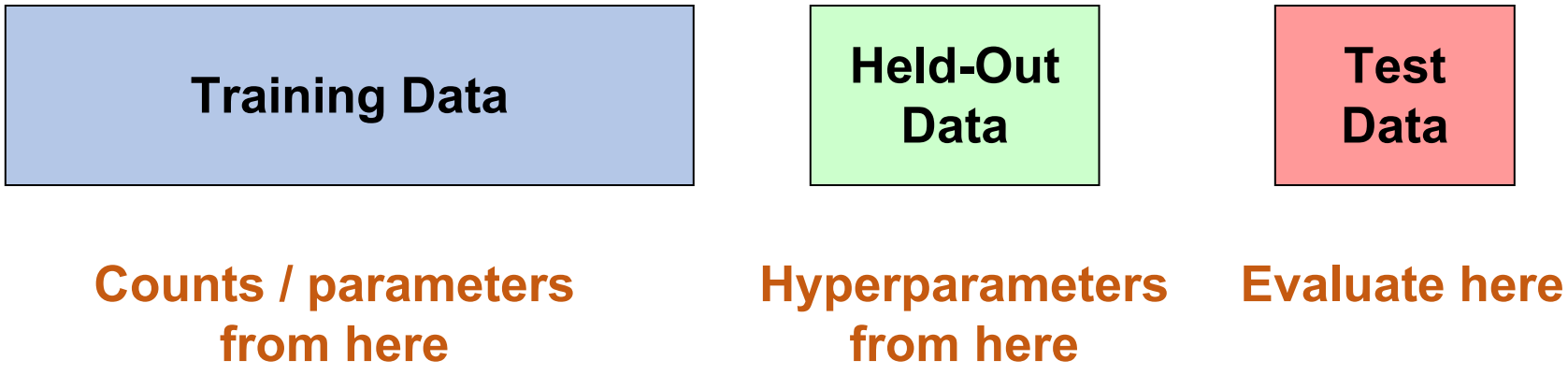
$$\log P(X|\theta) = \sum_{w \in X} \log P(w|\theta)$$

Perplexity: "average per word branching factor"

$$\text{perp}(X, \theta) = \exp\left(-\frac{\log P(X|\theta)}{|X|}\right)$$

Want to maximize likelihood on test, not training data

- Empirical n-grams won't generalize well
- Models derived directly from counts require smoothing parameters to be tuned on held-out data to simulate test generalization



- Set hyperparameters to maximize the likelihood of the held-out data